

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This automated official refresh report closes `wave 5.2B offset and harmonic guided refresh`.

Decision

- `wave52b_offset_harmonic_guided_registry` is closed as pending operator review; do not promote from scalar metrics alone.
- The strongest refreshed aggregate candidate is `wave52b_offset_centered_shape_harmonic_Fw`.
- The accepted direction-parallel baseline changes only after a human closure review records that promotion explicitly.
- This launcher-generated report is part of the same operator run as the matrix, collage, overlay, and PDF exports.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary:
`output/validation_checks/track2_reference_comparison/2026-07-02-12-43-56__track2_full_directional_family_matrix_wave52b_offset_harmonic_guided_track2_refresh_2026_07_02/validation_summary.yaml` ;
- per-condition metrics:
`output/validation_checks/track2_reference_comparison/2026-07-02-12-43-56__track2_full_directional_family_matrix_wave52b_offset_harmonic_guided_track2_refresh_2026_07_02/per_condition_metrics.csv` ;
- best-model collage report:
`doc/reports/analysis/track2/best_model_collage_report/[2026-07-02]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-07-02]/track2_multi_model_curve_comparison_report.md` ;
- operator launch logs:
`output/validation_checks/track2_operator_launch_logs/2026-07-02-12-43-48_wave52b_offset_harmonic_guided_track2_refresh_2026_07_02` .

Candidate Refresh

The refresh added 3 candidates from wave52b_offset_harmonic_guided_registry into the official 168 - candidate matrix.

Surface	Candidate	Family
global	wave52b_offset_centered_shape_harmonic_global	wave52b_offset_harmonic_guided
Fw	wave52b_offset_centered_shape_harmonic_Fw	wave52b_offset_harmonic_guided
Bw	wave52b_offset_centered_shape_harmonic_Bw	wave52b_offset_harmonic_guided

Refreshed Source Leaders

The table ranks the refreshed source by aggregate offline TE Curve Verification Pipeline metrics.

Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	wave52b_offset_centered_shape_harmonic_Fw	0.001695	0.002045	3.391	8.270
Bw	wave52b_offset_centered_shape_harmonic_Bw	0.002266	0.002708	3.986	9.758
global	wave52b_offset_centered_shape_harmonic_global	0.002221	0.002629	4.184	9.818

Refreshed Source Leaderboard

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
1	Fw	wave52b_offset_centered_shape_harmonic_Fw	0.001695	0.002045	3.391	8.270
2	global	wave52b_offset_centered_shape_harmonic_global	0.002221	0.002629	4.184	9.818
3	Bw	wave52b_offset_centered_shape_harmonic_Bw	0.002266	0.002708	3.986	9.758

Current Direction Leaders

These leaders are read from the matrix direction breakdown after the refresh.

Direction	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	track2g_curve_aware_full_curve_composite_global	0.002258	0.002737	4.040	10.132

Direction	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	rcim_retuned_ GBM19_Fw	0.001089	0.001299	2.372	4.912

Visual Evidence

The same launcher run regenerated the visual companion reports and verified that the refreshed source appears in the visual package.

Source	Collage	Overlay Forward	Overlay Backward
wave52b_offset_harmo nic_guided_registry	3	1	1

Closeout Decision

Wave 5.2B offset and harmonic guided refresh is closed as: pending operator review; do not promote from scalar metrics alone.

Use the Wave 5.2B curve-first evidence to decide whether harmonic-guided offset and centered-shape structure should enter Wave 5.2C or Wave 6 integration.